

Weekly Research Update

Three Project Progress Report

Muting (Don) Ma
September 2, 2026

Embedding Retrieval Capacity

Bi-objective Trajectory Revision C1

Stochastic Multiple Knapsack Manuscript

Public artifacts: [OpenProjects repository](#)

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1. Embedding Retrieval Capacity

Name: Muting (Don) Ma

Project: Embedding Retrieval Capacity

Date: September 2, 2026

1. Current objective

- Translate fixed-dimensional retrieval limits into a proposal-ready account of when representation error can undermine optimization decisions.

2. Progress since the last meeting

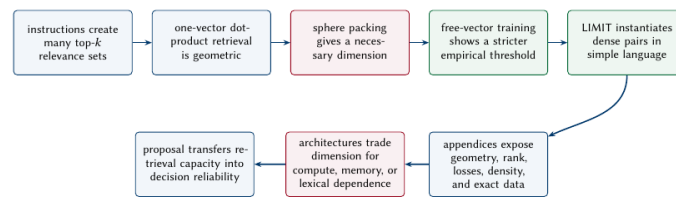
- **Completed:** Built a 60-frame, 72-slide dissection with prerequisites, the full proof, experiments, and every appendix.
- **Completed:** Separated the proved margin bound, the observed free-vector threshold, the LIMIT benchmark, and the architecture implications.
- **In progress:** Identified a distribution-dependent theorem that carries retrieval error into feasibility loss or operational regret as the proposal transfer.

3. Key result or lesson

- If every size- k answer must be realized with margin 2γ , then $d \geq \log \binom{n}{k} / \log(1 + 1/\gamma)$ [1].
- LIMIT shows a further gap between geometric capacity and learned use of that capacity.
- The **public causal map** connects both findings to decision reliability.

The Paper from Beginning to End

One connected causal map



Takeaway. The theorem establishes the floor, the experiments reveal additional gaps, and the appendices explain why precision, loss, and qrel structure matter.

On the Theoretical Limitations of Embedding-Based Retrieval

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Figure 1: Evidence chain from retrieval capacity to decision reliability.

4. Input needed from Dr. Gong

- Should the transfer theorem target feasibility loss, objective regret, or both?
- Should the first stress test compare single-vector retrieval with multi-vector retrieval, hybrid retrieval, or both?

5. Before the next meeting

- Draft the proposal subsection.
- Formalize the first error-to-regret proposition.
- Specify a toy operational stress test and metrics.

Public artifacts: [project folder](#); [selected snapshot](#).

2. Bi-objective Trajectory Revision C1

Name: Muting (Don) Ma

Project: Bi-objective Trajectory Revision C1

Date: September 2, 2026

1. Current objective

- Close revision C1 by making pair-pass uncertainty auditable as expected collision cost and completing the validation and comparison sequence.

2. Progress since the last meeting

- **Completed:** The revision/c1 branch derived the Rayleigh survival map and least-gap nested-event result, then removed the retired shared-cell average and count detour.
- **Completed:** The revised estimand regenerated schedule costs, the control menu, break-even results, figures, and the evidence boundary.
- **In progress:** The main closure audit records only Step 1 complete, with eight of nine plan steps still requiring work, so no merge was attempted.

3. Key result or lesson

- The schedule estimand is $C^{\text{cross}} = \kappa_c \sum_{\{q, q'\}} p(\gamma_{\min}(q, q'))$.
- Nested pair-pass events justify the least gap, and linearity permits the schedule sum without pair independence.
- The within-instance decision survives, but absolute cost remains model conditional until calibration.

The Final Answer in One View

Average or least probability

Do not use the retired average when the stated object is

a pair-pass probability or expected collision cost.

Use the least entry gap when the stated object is

the probability of at least one pass event under nesting.

Carry the expected monetary loss as

$$C^{\text{cross}} = \kappa_c \sum_{\{q, q'\}} p(\gamma_{\min}(q, q')).$$

Crossing Exposure Derivation

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Figure 2: Adopted least-gap expected collision cost construction.

4. Input needed from Dr. Gong

- Should the one-draw least-gap model remain the primary estimand while calibration is developed?
- Should the next work prioritize the audit and calibration package or practical-controller pilots?

5. Before the next meeting

- Export canonical pair records and the pair-level audit.
- Run the prescribed error-scale sensitivity.
- Specify controller comparisons and seed and demand pilots.

Public artifacts: project folder; selected snapshot.

3. Stochastic Multiple Knapsack Manuscript

Name: Muting (Don) Ma

Project: Stochastic Multiple Knapsack Manuscript

Date: September 2, 2026

1. Current objective

- Rebuild the stochastic multiple knapsack manuscript around a defensible contribution, a valid integer policy, and executable benchmarks.

2. Progress since the last meeting

- **Completed:** Built a 55-frame dissection with 62 logical slides that integrates all ten advisor points and every computational result.
- **Completed:** Reviewed four close papers that separate human and stochastic evidence from structural and computational evidence [2, 3, 4, 5].
- **Blocked:** The publication claim remains unsupported because the displayed feasibility proposition is false and the reported assignments come from the continuous relaxation.

3. Key result or lesson

- The defensible core is a time-discounted two-stage formulation plus a continuous Benders proof of concept.
- Zero effort and zero completion contradict the current feasibility proposition, while 12 of 21 reported assignments are fractional.
- The **public verdict** states the resulting revision boundary.

Final Verdict

Promising core, incomplete paper

What is worth keeping. The timing mechanism, scenario formulation, completion discounting, and clean LP decomposition provide a useful technical core.

What the four papers add. They convert a broad novelty claim into three measurable obligations: stochastic value, human outcome value, and algorithmic value.

What blocks publication. The current human-centered label is not modeled, one proposition is false as stated, and the main result tables interpret fractional LP variables as integer decisions.

Editorial judgment. The manuscript should be rebuilt around a narrower claim and executable benchmark comparisons before resubmission.

Takeaway. The paper has a coherent modeling seed, but its contribution, method, experiment, and interpretation do not yet meet at the same level of evidence.

Stochastic Multiple Knapsack Manuscript: Beginning-to-End Dissection

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Figure 3: Current technical and editorial verdict.

4. Input needed from Dr. Gong

- Should the paper narrow to managerial task scheduling or add an explicit individual outcome and fairness treatment?
- Should the first integer baseline use direct MILP, integer Benders, or rounding and repair?

5. Before the next meeting

- Correct the proposition and prove nontriviality.
- Produce feasible integer incumbents and gaps.
- Prespecify held-out tests, seeds, scaling, and ablations.

Public artifacts: project folder; selected snapshot.

References

- [1] O. Weller, M. Boratko, I. Naim, and J. Lee, “On the Theoretical Limitations of Embedding-Based Retrieval,” International Conference on Learning Representations, 2026. [ICLR paper page](#).
- [2] E. Gülmez, M. E. Aydin, K. B. Urganci, S. C. Gamoura, and H. I. Korucu, “Healthcare Staff Scheduling with Work-Life Balance Constraint Using Multi Objective Evolutionary Algorithms,” *Computers and Industrial Engineering*, vol. 220, 2026. doi:10.1016/j.cie.2026.112287.
- [3] C. Perreault-Lafleur, M. Carvalho, and G. Desaulniers, “A Stochastic Integer Programming Approach to Reserve Staff Scheduling with Preferences,” *International Transactions in Operational Research*, vol. 32, no. 1, pp. 289 to 313, 2025. doi:10.1111/itor.13298.
- [4] S. Mancini, C. Meloni, and M. Ciavotta, “A Decomposition Approach for Multidimensional Knapsacks with Family-Split Penalties,” *International Transactions in Operational Research*, vol. 31, no. 4, pp. 2247 to 2271, 2024. doi:10.1111/itor.13207.
- [5] M. Masmoudi, Y. Adouani, and B. Jarboui, “LP Relaxation and Dynamic Programming Enhancing VNS for the Multiple Knapsack Problem with Setup,” *International Transactions in Operational Research*, vol. 31, no. 3, pp. 1890 to 1916, 2024. doi:10.1111/itor.13213.